

EXPERIMENT NO: 13

Determination and plotting of intrinsic impedance and phase velocity of various medium Aim:

To determine and plot intrinsic impedance and phase velocity of various medium. **Software required:**

- SCILAB version 6.0.1 **Program:**

```
EXP 13.sci (C:\Users\user\OneDrive\Documents\EXP 13.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 13.sci (C:\Users\user\OneDrive\Documents\EXP 13.sci) - SciNotes
EXP 13.sci X
7
8 //Constants
9 uo = 4 * %pi * 10^(-7);
10 eo = 8.854 * 10^(-12);
11
12 // Intrinsic impedance
13 n = sqrt(uo * ur / (eo * er));
14
15 // Phase velocity
16 pv = 1 / sqrt(uo * ur * eo * er);
17
18 // Display results
19 mprintf("\nIntrinsic Impedance = %.6f Ohm\n", n);
20 mprintf("Phase Velocity = %.6e m/s\n", pv);
21
22 // Values for plotting
23 x = linspace(0, 10, 50);
24
25 y = n * ones(1, length(x));
26 z = pv * ones(1, length(x));
27
28 // Create figure
29 clf(1);
30 clf();
31
32 // Intrinsic impedance graph
33 subplot(2, 1, 1);
34 plot2d3(x, y);
35 xlabel("Distance");
36 ylabel("Intrinsic Impedance (Ohm)");
37 title("Intrinsic Impedance");
38
39 // Phase velocity graph
40 subplot(2, 1, 2);
41 plot2d3(x, z);
42 xlabel("Distance");
43 ylabel("Phase Velocity (m/s)");
44 title("Phase Velocity");
45
46 xgrid();
47
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\User\Documents\

Documents

My Music

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ex 6-5d

EXP 13.Sd~

Enter relative permeability μ_r : 3×10^{-6}

Enter relative permittivity ϵ_r : 6×10^{-6}

Intrinsic Impedance = 251.156206 Ohm

Phase Velocity = 8.327657e+06 m/s

-->

Variable Browser

Name	Value	Type	Visibility	Memory
lens	1x1	Graphic ...	local	216 B
leo	8.85e-12	Double	local	216 B
ler	54	Double	local	216 B
ln	251	Double	local	216 B
lv	8.33e+06	Double	local	216 B
lu	1.25e-05	Double	local	216 B
lv	24	Double	local	216 B
lx	1x50	Double	local	608 B
ly	1x50	Double	local	608 B

Command History

```
-- 10/08/2026 14:33:17 -- //
-- 10/08/2026 14:50:34 -- //
-- 24/08/2026 04:27:36 -- //
-- 24/08/2026 04:29:27 -- //
3*10^-7
6
4
2
5
5
3
3
4
```

News feed

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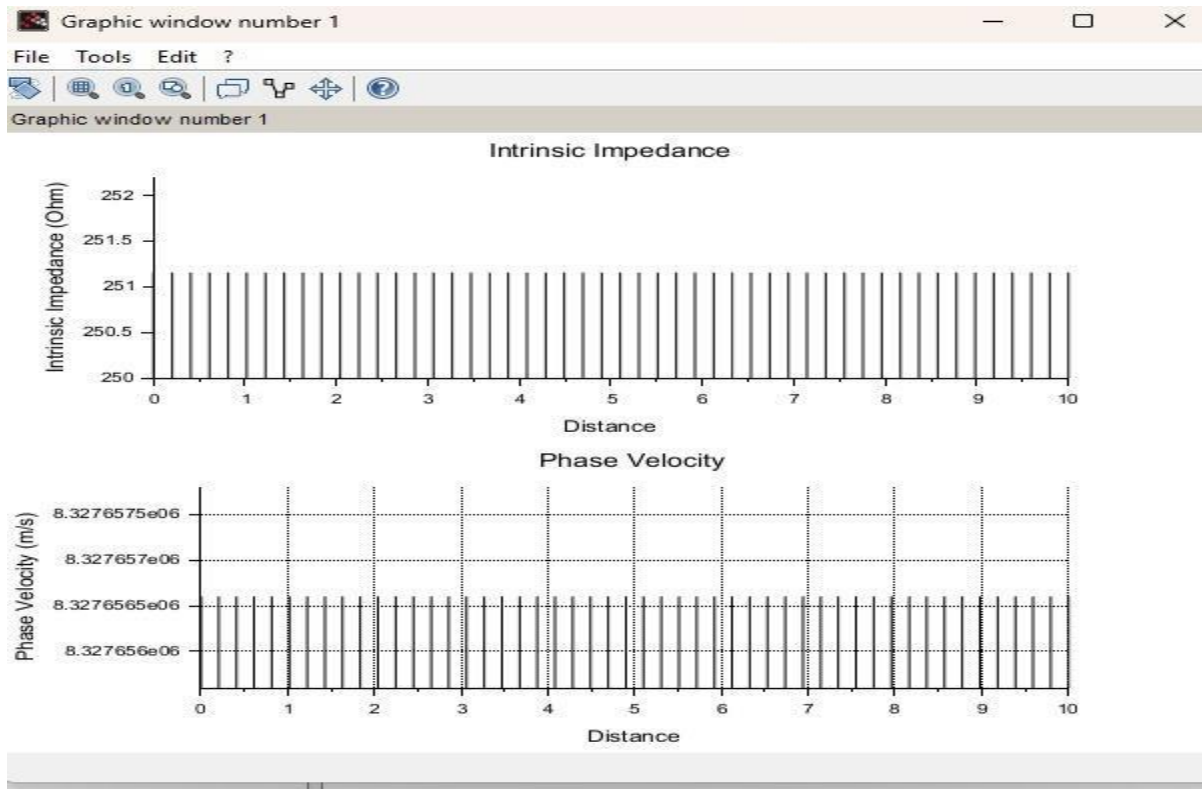
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- View and edit [help pages](#).
- Publish toolboxes on [ATOMS](#).
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ENG IN

5:26 AM 8/24/2026



Theoretical calculation:

$$\mu_r = 3 \times 10^{-4}$$

$$\epsilon_r = 4 \times 10^{-4}$$

$$\mu_0 = 4 \times 3.14 \times 10^{-7} = 1.256 \times 10^{-6}$$

$$\epsilon_0 = 8.854 \times 10^{-12}$$

Phase Velocity (pv)

$$pv = \frac{1}{\sqrt{\mu_0 \cdot \mu_r \cdot \epsilon_r \cdot \epsilon_0}}$$

$$\mu_0 \cdot \mu_r = (1.256 \times 10^{-6}) \times (3 \times 10^{-4}) = 3.768 \times 10^{-10}$$

$$\epsilon_0 \cdot \epsilon_r = (8.854 \times 10^{-12}) \times (4 \times 10^{-4}) = 3.5416 \times 10^{-15}$$

$$= (3.768 \times 10^{-10}) \times (3.5416 \times 10^{-15})$$

$$\Rightarrow 1.33447 \times 10^{-24}$$

$$= \sqrt{1.33447 \times 10^{-24}}$$

$$\Rightarrow 1.15519 \times 10^{-12}$$

$$pv = \frac{1}{1.15519 \times 10^{-12}}$$

$$= 8.6565 \times 10^{11}$$

Intrinsic Impedance (n)

$$n = \sqrt{\frac{\mu}{\epsilon}}$$

$$\frac{\mu_0 \cdot \mu_r}{\epsilon_0 \cdot \epsilon_r} = \frac{3.768 \times 10^{-10}}{3.5416 \times 10^{-15}} = 106392.59$$

$$n = \sqrt{106392.59} = 326.178 \Omega$$

$$= \frac{3.768 \times 10^{-10}}{4 \times 10^{-4}} \times (8.854 \times 10^{-12})$$

$$\Rightarrow 8.34047 \times 10^{-18}$$

$$n = \sqrt{8.34047 \times 10^{-18}}$$

$$= 2.8879 \times 10^{-9}$$

Case 1:

To determine the intrinsic impedance and phase velocity of electromagnetic waves in free space(vacuum).

Program:

Output :

Case 2:

The screenshot shows the Scilab 2026.1.0 Console window. The main area displays the following results:

```
Intrinsic Impedance (eta0) = 376.734309 Ohms  
Phase Velocity (vp) = 2.997956e+08 m/s
```

Below the results, there is a prompt: `--> |`

The Variable Browser on the right shows the following variables:

Name	Value	Type	Visibility	Memory
epsilon0	8.85e-12	Double	local	216 B
eta0	377	Double	local	216 B
mu0	1.26e-06	Double	local	216 B
vp	3e+08	Double	local	216 B

The Command History on the right shows the following commands:

```
-- // -- 10/08/2026 14:33:17 -- //  
-- // -- 10/08/2026 14:50:34 -- //  
-- // -- 24/08/2026 04:27:36 -- //  
-- // -- 24/08/2026 04:29:27 -- //  
-- 3*10^-7  
4  
4  
2  
5  
3  
4  
5  
3  
4
```

The News feed on the right shows a message from the Scilab community:

Scilab is open source.
Giving back support or use cases is appreciated.

- Add and improve functions on [Scilab](#).
- View and edit [help pages](#).
- Publish toolboxes on [ATOMS](#).
- Post questions on [Discourse](#).

To determine the intrinsic impedance and phase velocity for a wave propagating through a conductive medium like copper.

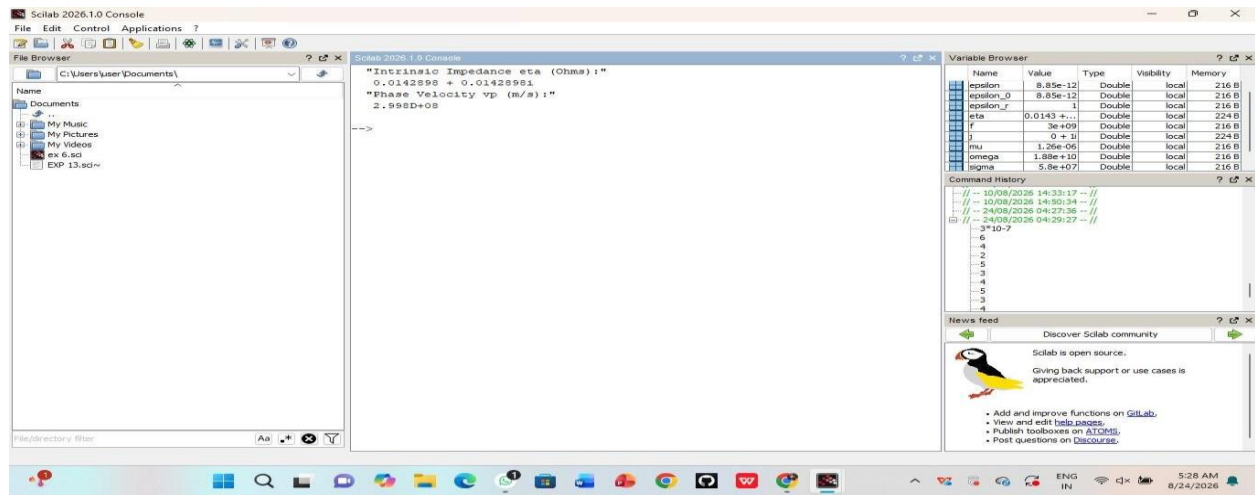
Program:

The screenshot shows the SciNotes editor with the following code:

```
1 clc;  
2 clear;  
3  
4 // Constants  
5 f = 3e9; // Frequency = 3 GHz  
6 mu = 4 * %pi * 10^(-7); // Permeability (H/m)  
7 sigma = 5.8e7; // Conductivity of copper (S/m)  
8 epsilon_0 = 8.854e-12; // Permittivity of free space (F/m)  
9 epsilon_r = 1; // Relative permittivity  
10  
11 // Imaginary unit  
12 j = %i;  
13  
14 // Effective permittivity  
15 epsilon = epsilon_0 * epsilon_r;  
16  
17 // Angular frequency  
18 omega = 2 * %pi * f;  
19  
20 // Intrinsic impedance  
21 eta = sqrt((j * omega * mu) / (sigma + j * omega * epsilon));  
22  
23 // Phase velocity  
24 vp = 1 / sqrt(mu * epsilon);  
25  
26 // Display results  
27 disp("Intrinsic Impedance-eta: (Ohms):");  
28 disp(eta);  
29  
30 disp("Phase Velocity-vp: (m/s):");  
31 disp(vp);  
32
```

Case 3:

Output :



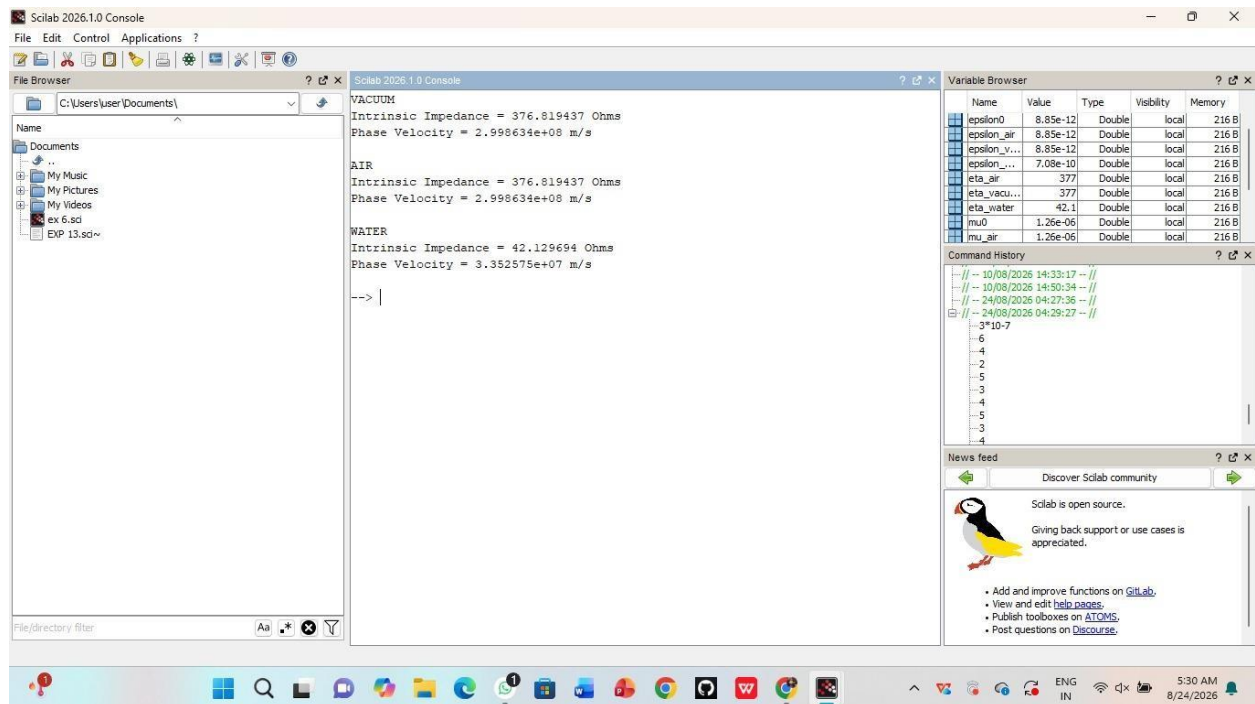
To determine and compare the intrinsic impedance and phase velocity of electromagnetic waves propagating through different media: Vacuum, Air, and Water.

Program:

Case 4:

```
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File Edit Format Options Window Execute ?
EXP 13.sci (C:\Users\user\OneDrive\Documents\EXP 13.sci) - SciNotes
EXP 13.sci X
1 clc;
2 clear;
3
4 // Constants
5 mu0 = 4 * %pi * 10^(-7); // Permeability of free space (H/m)
6 epsilon0 = 8.85 * 10^(-12); // Permittivity of free space (F/m)
7
8 // =====
9 // VACUUM
10 // =====
11 mu_vacuum = mu0;
12 epsilon_vacuum = epsilon0;
13
14 // =====
15 // AIR
16 // =====
17 mu_air = mu0;
18 epsilon_air = epsilon0;
19
20 // =====
21 // WATER
22 // =====
23 mu_water = mu0;
24 epsilon_water = 80 * epsilon0;
25
26 // Function to calculate properties
27 function [eta, vp] = calculate_properties(mu, epsilon)
28     eta = sqrt(mu / epsilon); // Intrinsic Impedance (Ohm)
29     vp = 1 / sqrt(mu * epsilon); // Phase Velocity (m/s)
30 endfunction
31
32 // =====
33 // VACUUM
34 // =====
35 [eta_vacuum, vp_vacuum] = calculate_properties(mu_vacuum, epsilon_vacuum);
36
37 fprintf("VACUUM\n");
38 fprintf("Intrinsic Impedance = %.6f Ohms\n", eta_vacuum);
39 fprintf("Phase Velocity = %.6e m/s\n", vp_vacuum);
40
41 // =====
42 // AIR
43 // =====
44 [eta_air, vp_air] = calculate_properties(mu_air, epsilon_air);
45
46 // =====
47 // WATER
48 // =====
49 [eta_water, vp_water] = calculate_properties(mu_water, epsilon_water);
50
51 fprintf("WATER\n");
52 fprintf("Intrinsic Impedance = %.6f Ohms\n", eta_water);
53 fprintf("Phase Velocity = %.6e m/s\n", vp_water);
```

Output:



Result:

Thus the determination and plotting of intrinsic and phase velocity is performed and successfully verified in copper.